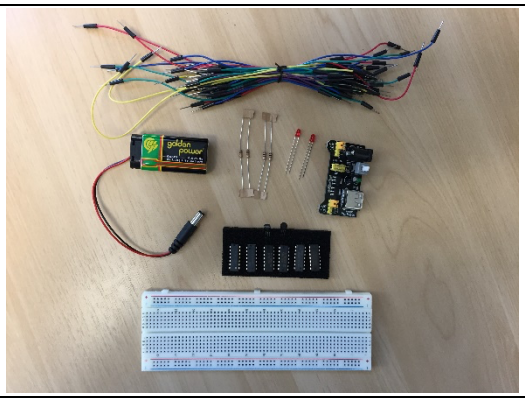
Lab 3

Objective: Build logic gates using transistors.

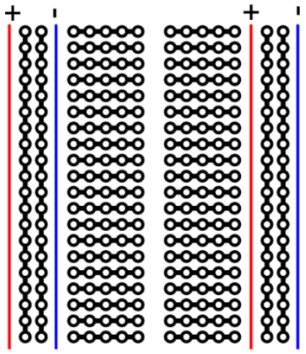
This lab builds on lecture 2

Each kit should contain:

2 * LED (light-emitting diode) 2 * 1K ohm resistor (Brown Black Red Gold) 2 * 10K ohm resistor (Brown Black Orange Gold) 1 * breadboard 1 * power board 1 * 9v battery (and connector) 2 * CTBC 547 JS Transistor (NPN) 2 * 74LS86 (quad 2-input XOR gate) 2 * 74LS08 (quad 2-input AND gate) 1 * 74LS00 (quad 2-input NAND gate) 1 * 74LS32 (quad 2-input OR gate) Bundle of wires	

The Breadboard

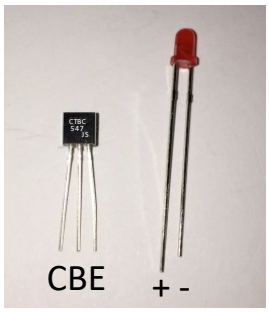
Breadboards are designed for experimenting with electronic circuits. They are a pre-printed set of connections inside a piece of moulded plastic designed to allow the plug-in of power, resistors, LEDs, chips, and so on. The breadboards we are using are connected like this:



From (http://www.backward-workshop.com/electronics/breadboard-curriculum/breadboard/)

LED and Transistor

The LED and Transistor are *polarised* so have a particular orientation. The LED is a diode, which allows electricity to flow only in one direction (from + to -). The Transistor allows flow from the collector (C) to the emitter (E) if current is applied to the base (B).



Units and Essential Formulae

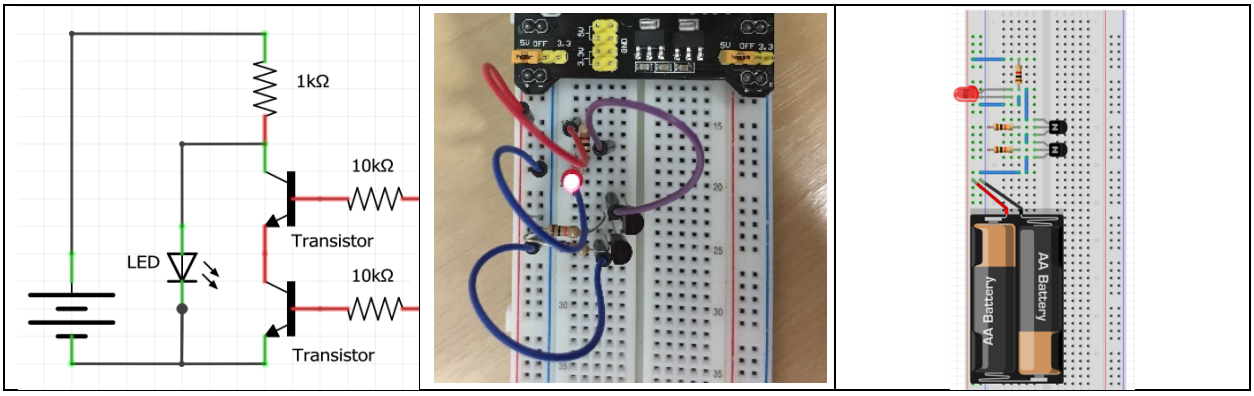
Quantity:	Symbol:	Unit:
Voltage	V	volt (V)
Current	I	amp (A)
Resistance	R	ohm (Ω)
Power	P	watt (W)

$$V = I \times R \qquad \text{(Ohm's Law)}$$

$$P = V \times I$$

Exercise 1: Two transistors make a NAND

In this exercise we're going to use two transistors to build a NAND gate. This is achieved by placing the transistors in series such that when they are both turned on they create a short circuit that turns off the LED. Once you've built the circuit, use the two base pins to show that you've built a NAND gate.



Exercise 2: One transistor makes a NOT gate!

Draw the circuit for a NOT gate (or look it up online). Build the circuit on the breadboard and show that you have, indeed, built a NOT gate.

Exercise 3: Two transistors make an OR gate

Draw the circuit diagram for the OR gate (or look it up online) then build the circuit on the breadboard. Show that you have, indeed, built an OR gate.

Summary

In this lab we built NAND, OR, and NOT gates from transistors. In the next lab we'll build a binary adder using logic chips (integrated circuits), which combine multiple gates into a single component.